Summary of agent reports on the MpT / holography paper

There are 12 reports: three models (Haiku, Sonnet, Opus 5.5), two effort levels (low, high), and two runs each.

Counts

Principal achievement. All 12 reports give the same answer: the near-horizon (Maldacena/IMSY) limit in the bulk is the *same* BPS/MpT decoupling limit that defines matrix theory at asymptotic infinity. This unifies matrix theory, AdS/CFT, non-Lorentzian holography and $T\bar{T}$. Reports differ mainly in how much weight they give the p-brane $T\bar{T}$ flow equations. Six reports call them the most concrete or checkable result (Opus high 1, Opus high 2, Sonnet high 1, Sonnet high 2, Sonnet low 2, Haiku high 2). Three reports checked the flow equation numerically: Haiku high 1 ($p = 1$), Opus high 1 ($p = 2$–5) and Opus high 2 ($p = 0$–5).

Principal drawback. The reports nearly agree here. All 12 note that the non-Lorentzian bulk geometries are not shown to solve the MpT supergravity equations of motion and torsional constraints. They differ in which point they lead with:

Headline drawback	Count
Claims are kinematic: the DLCQ^n/DLCQ^m conjecture is bookkeeping, and no observable is matched	8
New bulk geometries are unverified (no EOM check, ad hoc ℓ/N rescaling)	3
The "intrinsic" $T\bar{T}$ generation of the bulk is not established ($\Omega(r)$ is put in by hand, λ is position-dependent)	1

Next direction. The reports disagree most here.

Proposed follow-up	Count
Formulate MpT supergravity and verify the backgrounds (plus isometries)	6
Quantitative test of one DLCQ^1/DLCQ^2 pair (symmetries, protected spectrum)	3
Isometry algebra of a single bulk geometry vs. boundary symmetries	1
Position-dependent $T\bar{T}$ deformation at $p = 1$ (F1/NS5)	1
Non-abelian (and supersymmetric) p-brane $T\bar{T}$ flow	1

The non-abelian flow is the usual runner-up: about eight reports describe it as higher impact but lower feasibility.

Breakdown by model

Next direction	Haiku	Sonnet	Opus
MpT supergravity / verify backgrounds	2	4	0
Test a specific DLCQ^1/DLCQ^2 pair	0	0	3
Isometry algebra only	1	0	0
Position-dependent $T\bar{T}$ at $p = 1$	1	0	0
Non-abelian $T\bar{T}$ flow	0	0	1
Drawback: kinematic / non-predictive	1	3	4
Drawback: unverified geometries	2	1	0
Drawback: $T\bar{T}$ generation not intrinsic	1	0	0

Sonnet gives the most uniform answers: all four reports propose building MpT supergravity. Opus is the only model that proposes testing a concrete pair with matched observables. Two of its reports choose the D4/M0T example, and Opus high 2 shows that it lifts exactly to null-compactified $\mathrm{AdS}_7 \times S^4$. Haiku's answers vary the most.

Per-report table

Model	Run	Drawback (headline)	Next direction
Haiku	low 1	DLCQ conjecture untested beyond known cases	Check M2/D2 rescaled geometries against MpT sugra EOM
Haiku	low 2	Geometries not checked against EOM	MpT sugra consistency check ($p = 2, 3$, bosonic)
Haiku	high 1	Geometries unverified; ad hoc ℓ rescaling	Isometry algebra of a bulk geometry vs. boundary symmetries
Haiku	high 2	Intrinsic $T\bar{T}$ bulk generation not established	Position-dependent $T\bar{T}$ at $p = 1$ for F1/NS5
Sonnet	low 1	Geometries not checked against EOM; no observables	Build M0T/M2T sugra, verify, isometries, finite T
Sonnet	low 2	Structural, not dynamical; conjecture non-predictive	Build M0T/M2T sugra, verify, isometries, SUSY
Sonnet	high 1	Largely kinematic; DLCQ conjecture is bookkeeping	MpT sugra EOM, test ansatz, isometries, finite T
Sonnet	high 2	Kinematic and classificatory; unverified	MpT sugra EOM, classify $\Omega(r)$ ansatz, isometries
Opus	low 1	Rests on kinematics; no observable matched	Test M2/ABJM DLCQ^1/DLCQ^2 pair (index, algebra)
Opus	low 2	Reorganisation of known dualities, non-predictive	Non-abelian/SUSY $T\bar{T}$ flow vs. non-abelian DBI
Opus	high 1	Kinematic; no observable on any new pair	D4/M0T pair in M0T sugra; symmetries, BPS spectrum
Opus	high 2	Counting limits; no prediction that can fail	D4/M0T via $\mathrm{AdS}_7 \times S^4$ null uplift; spectrum test